

Project Progress Report

Campus Maintenance & Complaint Management System

Name: Ahnaf Tajwar

Roll: 2207104

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1. Project Overview

A campus maintenance complaint management system built on **Oracle Database (PL/SQL)** with a basic **Node.js + HTML/CSS/JS** web interface. Students submit maintenance complaints by location and category; supervisors assign specialized workers; the database enforces SLA deadlines, logs status changes, detects chronic issues, and tracks worker performance.

2. Work Completed So Far

- **Database design** — 9 normalized tables (3NF), ERD, data dictionary, constraints, sequences
- **Seed data** — 14 users, 24 locations, 38 sample complaints with assignments and feedback
- **PL/SQL layer** — 7 triggers, 6 stored procedures, 4 functions, 5 analytical views
- **Advanced SQL** — SLA analysis, worker overload, monthly trends, chronic issue queries
- **Backend API** — Express server connected to Oracle via `oracledb`
- **Basic frontend** — Role-based login and separate portals for student, worker, admin/supervisor
- **Documentation** — Schema design, Oracle setup guide, demo SQL scripts

Current UI is functional but **basic**. Core focus so far has been database logic and end-to-end connectivity. Visual design and advanced features are planned next.

3. Database Schema (9 Tables)

Table	Purpose
USERS	Students, workers, supervisors, admins
LOCATIONS	Building, floor, room registry
COMPLAINTS	Core complaint records with SLA and status
WORKERS	Worker profiles and performance scores
ASSIGNMENTS	Worker-to-complaint mapping and repair cost
STATUS_LOG	Audit trail for every status change

FEEDBACK	Student ratings after resolution
CHRONIC_FLAGS	Recurring issues at same location
MAINTENANCE_REPORTS	Monthly aggregated statistics

4. PL/SQL Components

7 Triggers

6 Procedures

4 Functions

5 Views

Role Grants

- SLA auto-deadline (urgent 4h, medium 24h, low 72h)
- Status audit logging and worker availability toggling
- Performance score recalculation after feedback
- Chronic issue detection (3+ same category at one location)
- Procedures: assign worker, monthly report, SLA escalation

5. Web Application (4 Pages)

Page / Role	Function
Login	Role-based sign-in with demo accounts
Student Portal	Submit complaints, view status, submit feedback
Worker Portal	View assigned jobs, start and resolve tasks
Admin / Supervisor Dashboard	Overdue list, assign workers, worker stats, reports

6. Supervisor vs Admin

Both roles share the **same dashboard UI** in the current version. In the database, **supervisor** can assign workers, monitor complaints, and run escalation. **Admin** has broader access including full table control and all reports. Demo logins: `omar.s@campus.edu` (supervisor), `admin@campus.edu` (admin).

7. Tech Stack

Oracle XE

PL/SQL

SQL Developer

Node.js

Express

oracledb

HTML/CSS/JS

8. Planned Next Steps

- UI enhancement — modern responsive design, charts, better navigation
- Secure password-based authentication (replace email-only demo login)
- Backend validation, error handling, and API improvements

- Email/SMS alerts for SLA breaches and chronic issues
- Exportable PDF/Excel maintenance reports from the web app
- Mobile-friendly layout and additional user features

9. Screenshots — Current UI (Basic Prototype)

Screens below show the working prototype connected to Oracle. UI polish is a planned next phase.

Figure 1 — Login Page

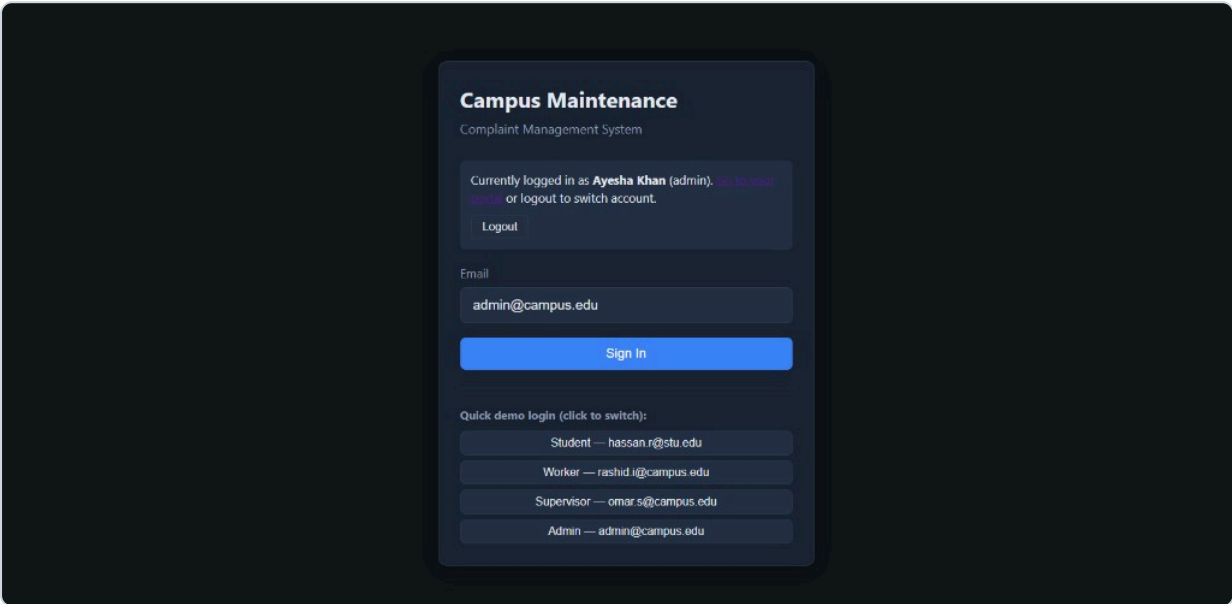


Figure 2 — Student Portal

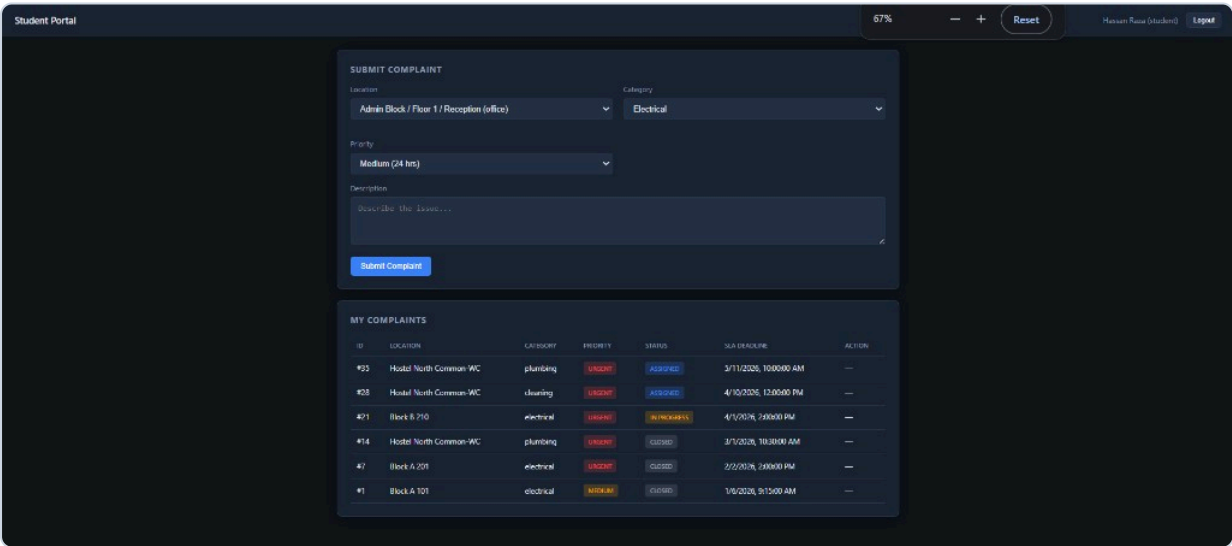


Figure 3 — Worker Portal

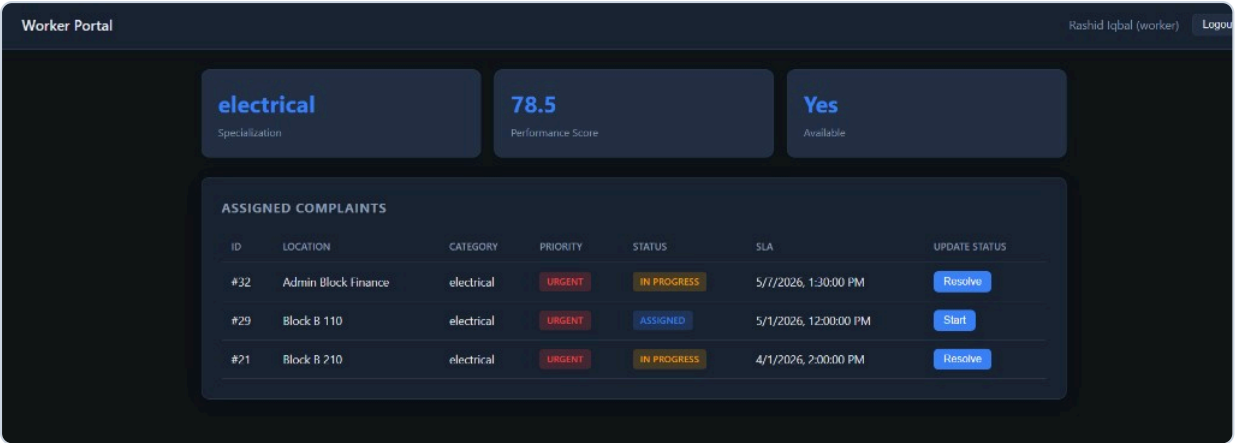


Figure 4 — Admin / Supervisor Dashboard

10

Overdue

2

Chronic Issues

4

Workers

5

Awaiting Assignment

ASSIGN WORKER

Complaint (submitted)

#37 — plumbing @ Hostel North Common-WC



Worker

Imran Baig (furniture)



Assign

GENERATE REPORT

Month

3

Year

2026

Generate

OVERDUE COMPLAINTS

ID	CATEGORY	PRIORITY	HOURS OVERDUE	LOCATION	WORKER
#24	plumbing	URGENT	1817.26h	Block B 010	Kamran Shah
#29	electrical	URGENT	1216.26h	Block B 110	Rashid Iqbal
#22	furniture	MEDIUM	1862.26h	Hostel North H-201	Imran Baig
#35	plumbing	URGENT	978.26h	Hostel North Common-WC	Kamran Shah
#32	electrical	URGENT	1070.76h	Admin Block Finance	Rashid Iqbal
#38	it	URGENT	927.76h	Block B 111	Nadia Farooq
#21	electrical	URGENT	1934.26h	Block B 210	Rashid Iqbal
#23	it	URGENT	1836.76h	Science Wing Lab-1	Nadia Farooq
#30	it	URGENT	1166.26h	Science Wing Lab-2	Nadia Farooq
#26	electrical	LOW	1699.26h	Block A 001	—

WORKER PERFORMANCE

WORKER	SPECIALIZATION	SCORE	OPEN	COMPLETED	OVERDUE	AVAILABLE
Rashid Iqbal	electrical	85	3	2	3	Yes
Nadia Farooq	it	93.33	3	3	3	Yes
Kamran Shah	plumbing	85	2	5	2	No
Imran Baig	furniture	80	1	3	1	Yes

CHRONIC ISSUES

LOCATION	CATEGORY	COUNT	FLAGGED
Hostel North / Common-WC	plumbing	5	3/11/2026, 8:00:00 PM
Science Wing / Lab-1	it	3	2/8/2026, 4:00:00 PM

MAINTENANCE REPORTS

MONTH/YEAR	TOTAL	RESOLVED	AVG HRS	TOTAL COST
3/2026	7	7	5.75	3500
2/2026	7	6	22.75	550
1/2026	6	5	18.5	5050

